Input-gradient space particle inference for neural network ensembles [1]

Trung Trinh[1], Markus Heinonen[1], Luigi Acerbi[2], Samuel Kaski[1,3] [2]
[1]Aalto University, [2]Helsinki University, [3]University of Manchester [3]
{trung.trinh, markus.o.heinonen, samuel.kaski}@aalto.fi, luigi.acerbi@helsinki.fi [4]

Overview [5]

TL;DR: We learn an ensemble of neural networks that is **diverse** with respect to their **input gradients**. [6]

Repulsive deep ensembles (RDEs) [1] [7]

Description: Train an ensemble $\{\boldsymbol{\theta}_i\}_{i=1}^M$ using Wasserstein gradient descent [2], which employs a kernelized repulsion term to diversify the particles to cover the Bayes posterior $p(\boldsymbol{\theta}|\mathcal{D})$ [8]

$$\boldsymbol{\theta}_i^{(t+1)} = \boldsymbol{\theta}_i^{(t)} + \eta_t \left(\underbrace{\nabla_{\boldsymbol{\theta}_i^{(t)}} \log p(\boldsymbol{\theta}_i^{(t)} \mid \mathcal{D})}_{\text{Driving force}} - \underbrace{\frac{\sum_{j=1}^N \nabla_{\boldsymbol{\theta}_i^{(t)}} k(\boldsymbol{\theta}_i^{(t)}, \boldsymbol{\theta}_j^{(t)})}{\sum_{j=1}^N k(\boldsymbol{\theta}_i^{(t)}, \boldsymbol{\theta}_j^{(t)})}}_{\text{Repulsion force}} \right)$$ [9]

- The driving force directs the particles towards high density regions of the posterior. [10]
- The repulsion force pushes the particles away from each other to enforce diversity.

Problem: It is unclear how to define the repulsion term for neural networks. [11]

- weight-space repulsion is ineffective due to overparameterization; [12]
- function-space repulsion often results in underfitting.

Defining the input-gradient kernel k [13]

Given a base kernel κ, we define the kernel in the input-gradient space for a minibatch of training samples $\mathcal{B} = \{(\mathbf{x}_b, y_b)\}_{b=1}^B$ as follows: [14]

$$k(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) = \boxed{\frac{1}{B}\sum_{b=1}^{B}} \boxed{\kappa(\nabla_{\mathbf{x}_b} \mathbf{f}(\mathbf{x}_b; \boldsymbol{\theta}_i)_{y_b}, \nabla_{\mathbf{x}_b} \mathbf{f}(\mathbf{x}_b; \boldsymbol{\theta}_j)_{y_b})}$$ [15]

Take the average over all samples in the mini-batch.

Compare the input gradients of two particles with respect to the same input.

We choose the RBF kernel on a unit sphere as the base kernel κ: [16]

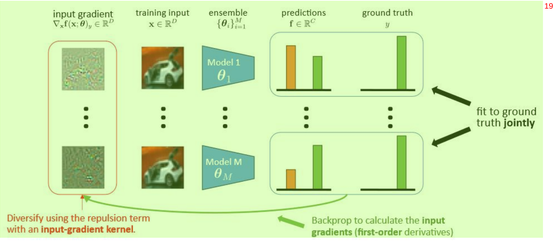
$$\kappa(\mathbf{s}_i, \mathbf{s}_j; \Sigma) = \boxed{\exp\left(-\frac{1}{\boxed{2h}}(\mathbf{s}_i - \mathbf{s}_j)^\top \boxed{\Sigma}^{-1}(\mathbf{s}_i - \mathbf{s}_j)\right)}, \quad \boxed{\mathbf{s}_i = \frac{\nabla_{\mathbf{x}} \mathbf{f}(\mathbf{x}; \boldsymbol{\theta}_i)_y}{||\nabla_{\mathbf{x}} \mathbf{f}(\mathbf{x}; \boldsymbol{\theta}_i)_y||_2}}$$ [17]

A scalar adaptively adjusted to prevent kernel vanishing.

Diagonal matrix containing the lengthscales.

Normalize input gradients to unit vectors.

First-order Repulsive deep ensembles (FoRDEs) [18]


 [19]

Possible advantages: [20]

- Each member is guaranteed to represent a different function; [21]
- The issues of weight- and function-space repulsion are avoided;
- Each member is encouraged to learn different features, which can improve robustness.

Main takeaways [26]

1. Input-gradient-space repulsion can perform better than weight- and function-space repulsion. [27]
2. Better corruption robustness can be achieved by configuring the repulsion kernel using the eigen-decomposition of the training data.

Benchmark comparison [28]

Table 1: **FoRDE-PCA achieves the best performance under corruptions while FoRDE-Identity outperforms baselines on clean data. FoRDE-Tuned outperforms baselines on both clean and corrupted data.** Results of RESNET18 / CIFAR-100 averaged over 5 seeds. Each ensemble has 10 members. cA, cNLL and cECE are accuracy, NLL, and ECE on CIFAR-100-C. [29]

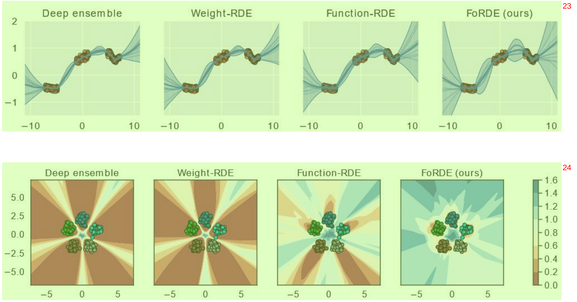
METHOD	NLL ↓	ACCURACY (%) ↑	ECE ↓	cA / cNLL / cECE
DEEP ENSEMBLES	**0.70**±0.00	81.8±0.2	0.041±0.003	54.3 / 1.99 / 0.05
WEIGHT-RDE	**0.70**±0.01	81.7±0.3	0.043±0.004	54.2 / 2.01 / 0.06
FUNCTION-RDE	0.76±0.02	80.1±0.4	0.042±0.005	51.9 / 2.08 / 0.07
FORDE-PCA (OURS)	0.71±0.00	81.4±0.2	0.039±0.002	**56.1 / 1.90 / 0.05**
FORDE-IDENTITY (OURS)	**0.70**±0.00	**82.1**±0.2	0.043±0.001	54.1 / 2.02 / 0.05
FORDE-TUNED (OURS)	**0.70**±0.00	**82.1**±0.2	0.044±0.002	55.3 / 1.94 / 0.05

Table 2: **FoRDE-PCA achieves the best performance under corruptions while FoRDE-Identity has the best NLL on clean data. FoRDE-Tuned outperforms most baselines on both clean and corrupted data.** Results of RESNET18 / CIFAR-10 averaged over 5 seeds. Each ensemble has 10 members. cA, cNLL and cECE are accuracy, NLL, and ECE on CIFAR-10-C. [31]

METHOD	NLL ↓	ACCURACY (%) ↑	ECE ↓	cA / cNLL / cECE
DEEP ENSEMBLES	0.117±0.001	96.3±0.1	0.005±0.001	78.1 / 0.78 / 0.08
WEIGHT-RDE	0.117±0.002	96.2±0.1	0.005±0.001	78.0 / 0.78 / 0.08
FUNCTION-RDE	0.128±0.001	95.8±0.2	0.006±0.001	77.1 / 0.81 / 0.08
FEATURE-RDE	0.116±0.001	96.4±0.1	**0.004**±0.001	78.1 / 0.77 / 0.08
FORDE-PCA (OURS)	0.125±0.001	96.1±0.1	0.006±0.001	**80.5 / 0.71 / 0.07**
FORDE-IDENTITY (OURS)	**0.113**±0.002	96.3±0.1	0.005±0.001	78.0 / 0.80 / 0.08
FORDE-TUNED (OURS)	0.114±0.002	96.4±0.1	0.005±0.001	79.1 / 0.74 / 0.07

References [33]
[1] F. D'Angelo and V. Fortuin, "Repulsive deep ensembles are Bayesian," Advances in Neural Information Processing Systems, vol. 34, pp. 3451–3465, 2021.
[2] C. Liu, J. Zhuo, P. Cheng, R. Zhang, and J. Zhu, "Understanding and Accelerating Particle-Based Variational Inference," in International Conference on Machine Learning, 2019.

Illustrative experiments [22]


 [23] [24]

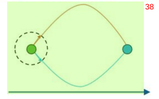
For a 1D regression task (above) and a 2D classification task (below), FoRDEs capture higher uncertainty than baselines in all regions outside of the training data. For the 2D classification task, we visualize the entropy of the predictive posteriors. [25]

Tuning the lengthscales Σ [34]

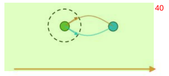
Each lengthscale is inversely proportional to the strength of the repulsion force in the corresponding input dimension. [35]

$$\underbrace{\frac{\partial}{\partial s_d}\kappa(\mathbf{s}, \mathbf{s}'; \Sigma)}_{\text{Repulsion force in the } d\text{-th dimension}} = -\frac{s_d - s'_d}{h\Sigma_{dd}}\kappa(\mathbf{s}, \mathbf{s}'; \Sigma) \propto \frac{1}{\underbrace{\Sigma_{dd}}_{\text{lengthscale in the } d\text{–th dimension}}}$$ [36]

Proposition: One should apply strong forces in high-variance dimensions (more in-between uncertainty) and weak forces in low-variance dimensions (less in-between uncertainty). [37]


[38]

In high-variance data dimensions, distances between data points are large, which lead to more in-between uncertainty → we can apply strong repulsion force to push the input gradients far away from each other. [39]


[40]

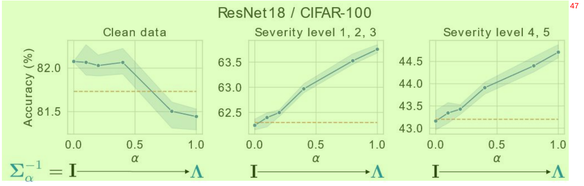
In low-variance data dimension, data points lie close to each other, leading to less in-between uncertainty → we need to use weak repulsion force. [41]

- Use PCA to get the eigenvalues and eigenvectors of the training data: $\{\mathbf{u}_d, \lambda_d\}_{d=1}^D$ [42]
- Define the base kernel:

$$\kappa_{\text{PCA}}(\mathbf{s}, \mathbf{s}'; \Sigma_\alpha) = \exp\left(-\frac{1}{2h}(\mathbf{U}^\top\mathbf{s} - \mathbf{U}^\top\mathbf{s}')^\top \Sigma_\alpha^{-1}(\mathbf{U}^\top\mathbf{s} - \mathbf{U}^\top\mathbf{s}')\right)$$ [43]

- $\mathbf{U} = [\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \cdots \;\; \mathbf{u}_D]$ is a matrix containing the eigenvectors as columns. [44]
- $\Sigma_\alpha^{-1} = (1-\alpha)\mathbf{I} + \alpha\boldsymbol{\Lambda}$ where $\boldsymbol{\Lambda}$ is a diagonal matrix containing the eigenvalues. [45]

Lengthscale tuning experiments [46]


 [47]

- Blue lines show accuracies of FoRDEs, while dotted orange lines show accuracies of Deep ensembles. [48]
- When moving from the identity lengthscale $\mathbf{I}$ to the PCA lengthscales $\boldsymbol{\Lambda}$:
  - FoRDEs exhibit small performance degradations on clean images of CIFAR-100;
  - while becomes more robust against the natural corruptions of CIFAR-100-C.